

# **LIVE EVENT DETECTION FOR PEOPLE'S SAFETY USING DEEP LEARNING**

## **ABSTRACT**

This project introduces a sophisticated real-time event detection system aimed at enhancing public safety measures. Leveraging advanced Computer Vision techniques and machine learning models, the system focuses on human pose estimation and activity recognition within live video streams. Utilizing the MediaPipe library, the system employs pre-trained machine learning models, likely based on convolutional neural networks, to accurately detect and track human body landmarks. The core algorithms include a Fall Detection Algorithm, which identifies potential falls by analyzing the positions of key body landmarks, and a Forbidden Zone Detection Algorithm, which monitors violations of predefined boundaries within the video frame. Additionally, the system calculates Frames Per Second (FPS), logs important events, records video streams, and provides email notifications for fall detections or forbidden zone violations. By seamlessly integrating these components, the system delivers timely alerts and actionable insights to authorities and responders, enabling swift responses to emerging situations and proactive measures to ensure public safety.